SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA1710**

COURSE OUTCOME COVERED

CO3: Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)

Assessment Tool Weightage: 20%

QUESTIONS: 5

Time Duration: 1 Hour
Total Marks:30

Annexure A

Resolution Algorithm Implementation

Code of Conduct:

(192524118)

I R. Akshaya *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: R. Akshaya

20+20+24+19+14

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Student 3: Akshaya R | Register Number: 192524118

Question 1 - Library Late Fine

Knowledge Base

1. If a book is returned late, then a fine is charged.
2. The book was returned late.

Goal

Prove that **a fine is charged.** using the Resolution Algorithm.

Tasks

1. Represent the statements in propositional logic.
2. Convert all statements into CNF. Negate the goal.
3. Apply the Resolution Algorithm step by step.
4. Show the Empty Clause (Box) if the goal is proved.

Question 2 - Flight Delay Notification

Knowledge Base

1. If a flight is delayed, then passengers are notified.
2. Flight 202 is delayed.

Goal

Prove that **passengers are notified.** using the Resolution Algorithm.

Tasks

1. Represent the statements in propositional logic.
2. Convert all statements into CNF. Negate the goal.
3. Apply the Resolution Algorithm step by step.
4. Show the Empty Clause (Box) if the goal is proved.

Question 3 - Employee Certification

Knowledge Base

1. If an employee completes the training, then the employee receives a certification.
2. Kevin completed the training.

Goal

Prove that **Kevin receives a certification.** using the Resolution Algorithm.

Tasks

1. Represent the statements in propositional logic.
2. Convert all statements into CNF. Negate the goal.
3. Apply the Resolution Algorithm step by step.
4. Show the Empty Clause (Box) if the goal is proved.

Question 4 - Medical Attention Alert

Knowledge Base

1. If a patient has a fever, then the patient needs medical attention.
2. Ravi has a fever.

Goal

Prove that **Ravi needs medical attention.** using the Resolution Algorithm.

Tasks

1. Represent the statements in propositional logic.
2. Convert all statements into CNF. Negate the goal.
3. Apply the Resolution Algorithm step by step.
4. Show the Empty Clause (Box) if the goal is proved.

Question 5 - Order Shipping Confirmation

Knowledge Base

1. If a payment is verified, then the order is confirmed.
2. If the order is confirmed, then the item is shipped.
3. The payment was verified.

Goal

Prove that **the item is shipped.** using the Resolution Algorithm.

Tasks

1. Represent the knowledge base in propositional logic.
2. Convert all implications into CNF.
3. Negate the goal.
4. Apply the Resolution Algorithm to derive the Empty Clause (Box).
5. Explain each resolution step and write the final conclusion.

1. LIBRARY LATE FINE:

1. Propositional Logic.

Let : L = Book was returned late.

F = Fine is charged.

Knowledge Base:

1. If a book is returned late, then a fine is charged.

$L \rightarrow F$

2. The book was returned late.

L

Goal : F

2. CNF and Negation of Goal.

Convert implication:

$L \rightarrow F \equiv \neg L \vee F$

Clauses:

$C_1 : \neg L \vee F$

$C_2 : L$

Negate the goal: $\neg F$

So, clauses are: $C_1 = \neg L \vee F$

$C_2 = L$

$C_3 = \neg F$

3. Resolution Algorithm :

Resolve C_1 and C_2 :

$(\neg L \vee F)$, (L)

Resolving $\neg L$ and L :

F

Now resolve with C_3 : (F), $(\neg F)$

$\therefore$ Empty clause.

4. Conclusion :

Since the Empty clause is obtained, the goal is proved.

$\therefore$ A fine is charged.

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2. FLIGHT DELAY NOTIFICATION :

1. Propositional Logic :

Let : D = Flight 202 is delayed
N = Passengers are notified.

Knowledge Base :

1. If a flight is delayed, then the passengers are notified.

$D \rightarrow N$

2. Flight 202 is delayed.
D

Goal : N.

2. CNF and Negation of Goal:

$D \rightarrow N \equiv \neg D \vee N$

Clauses: $C_1 : \neg D \vee N$

$C_2 : D$

$C_3 : \neg N \rightarrow$ Negated goal.

3. Resolution:

Resolve C_1 and C_2 : $(\neg D \vee N), (D)$

$\therefore N$

Resolve with C_3 : $(N), (\neg N)$

$\therefore$ Empty.

Passengers are notified.

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3. EMPLOYEE CERTIFICATION:

1. Propositional Logic:

Let T = Kevin completed the training

C = Kevin receives certification.

Knowledge base:

1. If an Employee completes training, then receives certification $\rightarrow T \rightarrow C$.

2. Kevin completed the training

T

Goal: C

2. <u>CNF and Negation of Goal:</u>

$T \to C \equiv \neg T \vee C$

Clauses: $C_1 : \neg T \vee C$

$C_2 : T$

$C_3 : \neg C$ ← Negated Goal.

3. <u>Resolution:</u>

Resolve C_1 and C_2 : $(\neg T \vee C), (T)$

$\therefore C$

Resolve C with C_3 : $(C), (\neg C)$

$\therefore$ (Empty).

Since the Empty clause is derived.

$\therefore$ Kevin receives a certification.

4. MEDICAL ATTENTION ALERT.

1. Propositional logic:

Let F = Ravi has a fever.

M = Ravi needs medical attention.

knowledge base:

1. If a patient has a fever, then the patient needs medical attention.

$F \rightarrow M$

2. Ravi has fever : F.

Goal : M.

2. CNF and Negation of Goal:

$F \rightarrow M \equiv \neg F \vee M$

Clauses : $C_1 : \neg F \vee M$, $C_2 : F$, $C_3 : \neg M$ ← Negated goal.

3. Resolution:

Resolve C_1 and C_2 : $(\neg F \vee M)$, (F)

∴ M

Resolve with C_3 : (M), $(\neg M)$.

∴ (Empty clause)

Since the Empty clause is obtained

∴ Ravi needs medical attention.

5. ORDER SHOPPING CONFIRMATION:

1. Propositional Logic:

Let S = Item is Shipped, C = Confirmed order,
P = Payment is verified.

Knowledge Base: 1. If payment is verified, then the order is confirmed. $P \rightarrow C$

2. If order is confirmed, then item is shipped. $C \rightarrow S$

3. Payment was verified: P

Goal: S.

2. Convert implications into CNF:

For statement 1: $P \rightarrow C$ becomes: $\neg P \vee C$

For statement 2: $C \rightarrow S$ becomes: $\neg C \vee S$

Fact: P, Negate the goal: $\neg S$

$\therefore$ Clauses are: $C_1: \neg P \vee C$, $C_2: \neg C \vee S$, $C_3: P$,
$C_4: \neg S$.

3. Resolution Algorithm:

Step 1: Resolve C_1 & C_3
$C_1: \neg P \vee C$, $C_3: P$
$\therefore C_1 + C_3 \rightarrow C$

Step 2: Resolve C_2.
$C_2: \neg C \vee S$, new clause: C
$\therefore C_2 + C \rightarrow S$.

While resolving S with negated goal,
Output: Empty clause $(S), (\neg S) \rightarrow (\ )$

$\therefore$ The Item is shipped.